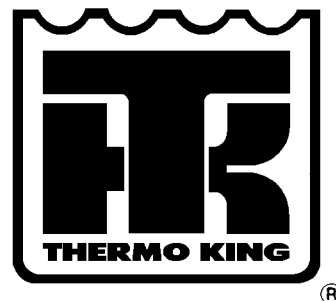




SERVICE BULLETIN



SUBJECT: μ P-D Controller Emergency Bypass Procedure

BULLETIN NO. C 024

DATE: August 7, 1998

Use the Emergency Bypass Switch to manually control unit operation if the μ P-D controller or output module fails.

NOTE: *If the output module fails, the unit will stop immediately. If the μ P-D controller fails but the output module is ok, the output module will:*

- *Immediately stop the unit on Fresh Loads,*
 - *Continue unit operation in Cool on Frozen Loads.*
- Use the Emergency Bypass Switch to take manual control of the unit.**

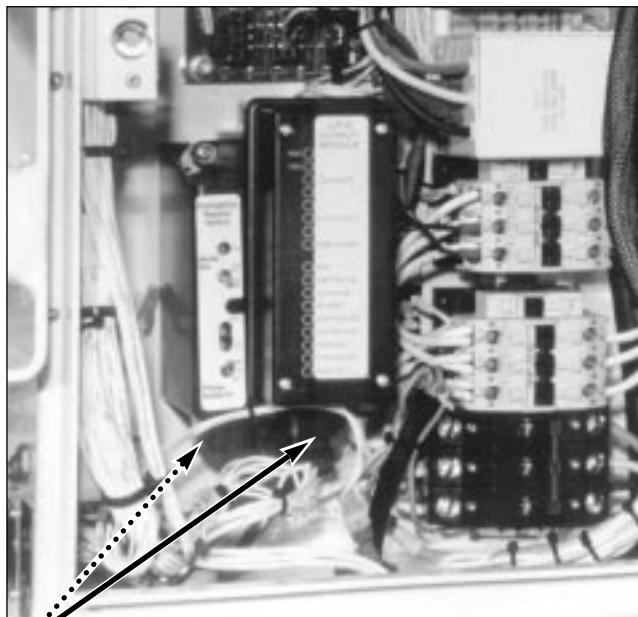
1. Turn the Unit On/Off switch to OFF.
2. Turn the power supply OFF. Disconnect the unit power cord from the power supply.

WARNING: *High voltage (460/380 volts) is present on the contactors and relays in the control box. To prevent dangerous electrical shock when working in the control box, disconnect the power supply to the unit whenever possible.*

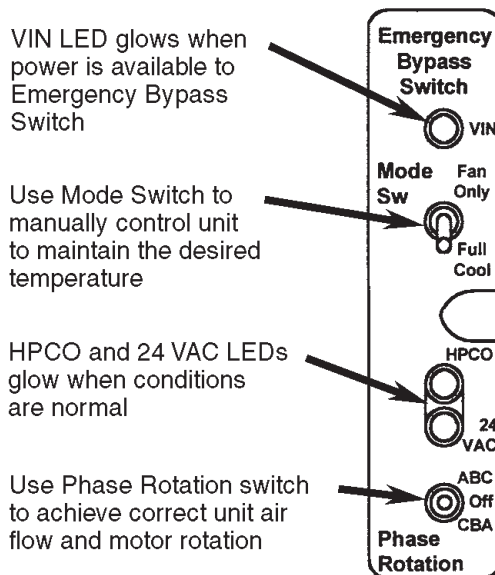
3. Disconnect wire harness connector "C1" from the bottom of the Output Module. Then connect "C1" to the bottom of the Emergency Bypass Switch assembly.

NOTE: *Make sure the Phase Rotation Switch on the Emergency Bypass Module is OFF.*

4. Connect the unit power cord to the proper power supply. Turn power supply ON.
5. Turn the Unit On/Off switch ON.
 - The VIN LED will glow to indicate power is available at the Emergency Bypass Switch.
6. Place the Phase Rotation Switch in the ABC position to start the unit.
7. Place the Mode Switch to Full Cool position to operate compressor and condenser fan.
 - Check condenser fan airflow. Air should be blowing out from the center of the grille.
 - If air is NOT blowing out from the center of the grille, place the Phase Rotation switch in the



Disconnect the C1 harness from bottom of the Output Module. Then connect it to the bottom of the Emergency Bypass Switch assembly.



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OFF position. Allow the motors and fans to come to a COMPLETE stop. Then place the Phase Rotation switch in the CBA position. Check condenser airflow again to confirm that air is blowing out from the center of the grille.

8. Use the Mode Switch to set unit operation for Evaporator Fan Only or Full Cool operation.

CAUTION: The unit does not control to setpoint during emergency bypass switch use. The unit must be cycled manually to maintain the desired temperature. Monitor the container temperature using the controller display (if working) or an external thermometer.

9. High pressure cutout (HPCO) and 24 Vac LEDs on the Emergency Bypass Module turn ON to indicate normal operation.
 - HPCO LED is ON when the HPCO switch is closed (normal). If the HPCO switch opens (high discharge pressure), the LED will turn OFF.
 - 24 VAC LED is ON when 24 volt ac control power is supplied to the Emergency Bypass Switch.

NOTE: If emergency bypass procedures are used under high ambient and high container (load) temperature conditions, a compressor motor shutdown may occur (indicating the compressor motor internal overload protector has tripped). Wait a few minutes to allow the overload protector to cool.

The unit is equipped with a suction modulation capacity control system to control system cooling capacity and limit compressor motor current draw. If the unit shuts down due to the overload protector tripping during emergency bypass procedure operation, restrict refrigerant flow to the compressor using the receiver tank outlet valve. Manually close the valve and then open the valve 1/4 turn. Restart the unit. Check compressor motor current draw. The motor should draw 14 to 15 amps. Adjust the receiver tank outlet valve slightly if necessary to raise or lower current draw to 14 to 15 amps.